ANALYTICAL CHEMISTRY CUMULATIVE EXAM

October 26, 1991

Below is a schematic of part of a radiochemical separation scheme for chemical analysis.

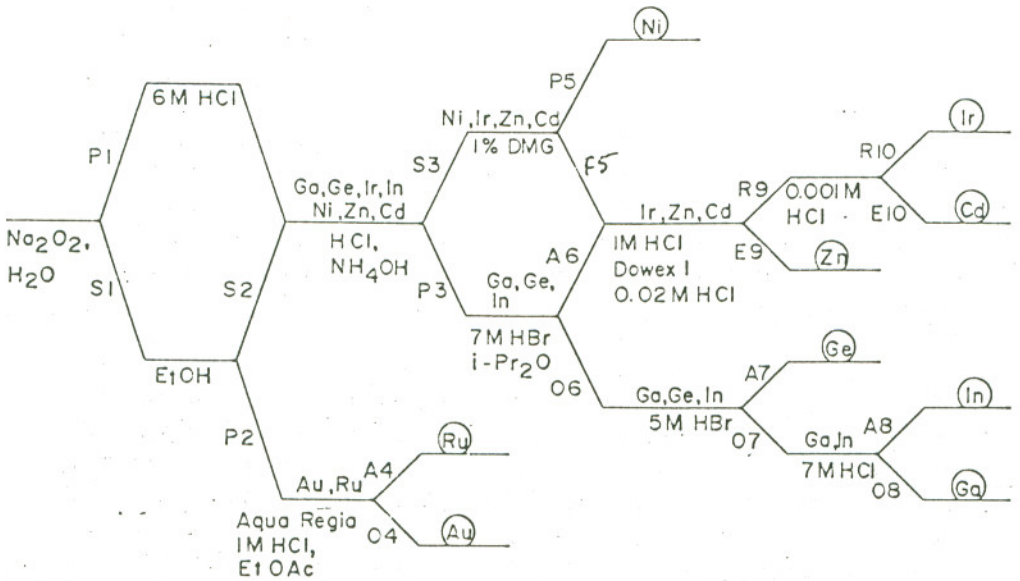

Fig. 1. Flow diagram of radiochemical separation procedure. Letters and numbers designate various phases

A aqueous layer
E eluate
O organic layer
P precipitate
R resin
S supernate

a) Locate points in the scheme where holdback carriers are used, state what these carriers might be and explain the reason for their use.

b) Ignoring decay in the reactor, what ^{60}Co activity would you expect to be induced in a sample which was 930 mg/g Co by a one hour irradiation in a reactor with a flux of 10^{14}n cm^{-2}s^{-1}. The relevant part of the Chart of the Nuclides is attached.

c) The Au and Ir isotopes used are γ emitters. Would you use a Ge(Li) or NaI detector for the counting? Give a detailed justification, bearing in mind that (n, γ) reactions are the most common reactions induced in the reactor.

d) In a certain experiment, the 1333 keV peak of ^{60}Co was found to contain 22,000 counts and the background 3500 counts, both over a 30 minute count time. Based solely on counting statistics, what is the uncertainty in the true (corrected for background) count rate at the 68% confidence level.